

UMGS: UAV Multispectral 3D Gaussian Splatting for Spectral-Index Synthesis

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UAV multispectral 3D reconstruction is difficult because multispectral bands are weak geometry drivers: they contain less texture than RGB, and UAV rigs introduce cross-FOV, resolution, and alignment differences. Direct spectral reconstruction can be unreliable: independently reconstructed bands may drift in geometry and support, preventing stable 3D spectral-index synthesis. We introduce UMGS, a two-stage 3D Gaussian Splatting framework that reconstructs an RGB Gaussian anchor and learns multispectral appearance with Gaussian positions, covariances, opacity, and identity frozen. This geometry lock lets spectral bands inherit stable 3D support while retaining band-specific appearance. We instantiate UMGS as independent per-band transfer (UMGS-I) and joint band-bank transfer (UMGS-J). Experiments on raw UAV and aligned aerial multispectral scenes show stable spectral renderings and index maps while preserving shared 3D support, and UMGS provides stronger product-oriented behavior than the available multispectral Gaussian baseline in official-average comparison and validation-gated reference-depth diagnostics.

CCS Concepts: • **Computing methodologies** → **Image-based rendering: Reconstruction**; • **Applied computing** → **Earth and atmospheric sciences**.

Additional Key Words and Phrases: multispectral imaging, 3D Gaussian Splatting, 3D reconstruction, UAV

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1 Introduction

UAV multispectral imaging is increasingly used in precision agriculture and environmental monitoring, where decisions are made from cross-band products rather than from a single rendered view [Colomina and Molina 2014; Mathews and Jensen 2013; Turner et al. 2015; Zhang and Kovacs 2012]. A reconstruction used for spectral-index synthesis must therefore satisfy two requirements at once: accurate per-band appearance and stable geometric correspondence across bands. The latter is essential because even small support drift can bias ratio-based measurements such as NDVI, GNDVI, and NDRE [Gitelson et al. 1996; Goetz et al. 1985; Huete 1988; Manolakis et al. 2016; Rouse et al. 1974].

Raw UAV multispectral data make this difficult. Multispectral bands are weak geometry drivers: they often have lower texture and less reliable local features than RGB. Practical UAV rigs add a second source of instability because RGB and multispectral cameras can differ in field of view, resolution, and spectral response. Under raw

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UAV multispectral captures, direct spectral reconstruction or independent per-band Gaussian optimization can therefore yield models whose geometry and support diverge across bands, preventing a stable unified 3D representation for index synthesis.

Recent radiance-field and Gaussian methods greatly improved training speed and view quality [Barron et al. 2022; Huang et al. 2024; Kerbl et al. 2023; Mildenhall et al. 2020; Müller et al. 2022; Yu et al. 2024]. Emerging multispectral, hyperspectral, and thermal Gaussian lines [Chen et al. 2024; Lu et al. 2024a; Meyer et al. 2025; Thirgood et al. 2025] further show that spectral rendering and index synthesis are feasible. UMGS addresses the UAV-specific geometry problem: how to exploit RGB as a reliable geometric carrier while learning low-texture spectral appearance on the same 3D support.

We introduce UMGS, a UAV multispectral 3D Gaussian Splatting framework for spectral-index synthesis. UMGS formulates multispectral reconstruction as *RGB-anchored geometry locking*: geometry, opacity, and Gaussian identity are learned once from RGB and then frozen during G/R/RE/NIR transfer. This design turns RGB into a stable geometric carrier, while each spectral band only learns appearance on the same splat support. We evaluate two implementations of the same constraint:

- **UMGS-I**: independent per-band transfer from a shared RGB anchor.
- **UMGS-J**: joint multispectral training with band-specific appearance banks in one unified checkpoint.

UMGS-I is our conservative default realization, while UMGS-J tests whether the same lock can be implemented as a joint bank representation. Both preserve identical support by construction; they differ only in how spectral appearance parameters are organized and optimized.

For raw UAV captures, alignment quality is another bottleneck. Cross-band FOV and resolution mismatch can make naive baselines unreliable, especially for NIR. We therefore use a baseline-aware rectification QA protocol as supporting infrastructure: it separates strict pass, pass-with-warning, and fail, and logs per-frame matching/stability diagnostics rather than relying on a single opaque gate.

Our main evaluation covers a 10-scene UAV-focused benchmark: 7 raw UAV captures and 3 aligned aerial multispectral scenes. The raw track validates end-to-end applicability (alignment, QA, reconstruction, and products). The aligned track removes rectification confounds and supports fair common-mask comparison with MS-Splatting. Across this suite, UMGS-I and UMGS-J are near-tied and consistently preserve shared support. Scratch-MS and Unfrozen-support are useful upper-bound ablations for single-band fitting, but they do not satisfy the shared-support requirement for stable multispectral index synthesis.

The contributions are:

- A geometry-locked multispectral Gaussian formulation that uses RGB as the geometric carrier for stable spectral-index synthesis.
- Two concrete UMGS realizations implementing the same lock with independent (UMGS-I) and joint (UMGS-J) spectral parameterizations.
- Unified image, index, cost, and common-mask reporting on a UAV-focused benchmark, with reference-depth diagnostics reported only when the reference source passes validation.
- Supporting raw-scene infrastructure for baseline-aware rectification QA with explicit diagnostic states and decision inputs.

2 Related Work

2.1 Radiance Fields and Gaussian Scene Representations

NeRF established neural volumetric rendering as a practical paradigm for novel view synthesis [Mildenhall et al. 2019, 2020]. Follow-up work improved anti-aliasing, unbounded scenes, and training speed, including mip-NeRF/mip-NeRF360, Instant-NGP, Plenoxels, TensorRF, and DVGO [Barron et al. 2022, 2021; Chen et al. 2022; Fridovich-Keil et al. 2022; Muller et al. 2022; Sun et al. 2022]. Large-scene and explicit-factorized lines such as NSVF, K-Planes, Zip-NeRF, MERF, Mega-NeRF, and Block-NeRF further explored memory/scale trade-offs [Barron et al. 2023; Fridovich-Keil et al. 2023; Liu et al. 2020; Reiser et al. 2023; Tancik et al. 2022; Turki et al. 2022].

3D Gaussian Splatting (3DGS) moved from volumetric integration to explicit anisotropic splats, enabling faster optimization and real-time rendering [Kerbl et al. 2023]. Subsequent work improved alias control, structural priors, compactness, and scene scale, including Mip-Splatting, Scaffold-GS, 2DGS, SuGaR, LightGaussian, Compact3DGS, Dynamic 3DGS, CityGaussian, and CityGaussianV2 [Fan et al. 2024; Guédon and Lepetit 2024; Huang et al. 2024; Lee et al. 2024; Liu et al. 2024, 2025; Lu et al. 2024b; Luiten et al. 2024; Yu et al. 2024]. These methods target rendering quality, memory, or geometry fidelity in RGB-dominant settings. UMGS addresses a different constraint: preserving one stable 3D support while transferring low-texture UAV spectral appearance onto it.

2.2 Multispectral and Thermal Gaussian Reconstruction

Multispectral remote sensing has long emphasized calibration, cross-band consistency, and physically meaningful indices [Gitelson et al. 1996; Goetz et al. 1985; Huete 1988; Manolakis et al. 2016; Rouse et al. 1974]. In UAV workflows, geometric preprocessing and modality alignment are often the limiting factor for downstream reliability [Colomina and Molina 2014; Mathews and Jensen 2013; Turner et al. 2015; Zhang and Kovacs 2012]. While deep learning for spectral imagery is mature in classification/segmentation tasks [Audebert et al. 2019; Li et al. 2019; Paoletti et al. 2019; Zhong et al. 2018], 3D multispectral scene reconstruction is still emerging.

Recent Gaussian-based spectral works include MS-Splatting for multi-camera multispectral NVS, HyperGS for hyperspectral latent rendering, MSGS for wavelength-aware Gaussian modeling, and thermal-domain lines such as ThermalGaussian and Thermal3D-GS [Chen et al. 2024; Lu et al. 2024a; Meyer et al. 2025; Thirgood et al. 2025; Zhang et al. 2026]. These works confirm the feasibility of

spectral Gaussian rendering and, in several cases, already store multiple spectral or modal signals in one Gaussian-based representation. MS-Splatting, for example, learns compact per-splat features and decodes spectral values with an MLP; HyperGS performs hyperspectral rendering in a learned latent space; ThermalGaussian learns registered RGB/thermal Gaussians after camera calibration. Our focus is complementary: raw UAV multispectral deployment, where spectral bands are weak geometry drivers and RGB/MS cameras can differ in FOV, resolution, and alignment. Rather than relying on each spectral band to define its own geometry, UMGS uses RGB as the geometric carrier and learns spectral appearance on the locked support.

2.3 SfM, Matching, and Cross-Modal Alignment

Practical reconstruction still depends on robust SfM/MVS foundations [Fischler and Bolles 1981; Hartley and Zisserman 2004; Schönbberger and Frahm 2016; Schönbberger et al. 2016]. Hand-crafted and learned feature matching both remain relevant, from SIFT/SURF/ORB and ECC-style alignment to SuperPoint, SuperGlue, LoFTR, LightGlue, and RoMa [Bay et al. 2008; DeTone et al. 2018; Edstedt et al. 2024; Evangelidis and Psarakis 2008; Lindenberger et al. 2023; Lowe 2004; Rublee et al. 2011; Sarlin et al. 2020; Sun et al. 2021]. Robust estimators (LO-RANSAC, USAC, MAGSAC) are essential under modality gaps and outlier-heavy correspondences [Barath et al. 2019; Chum et al. 2003; Raghunandan et al. 2013].

Our raw-scene QA protocol follows this systems view: it reports match quality and geometric stability separately, preserves legacy strict criteria for strong baselines, and uses a selective warning path only for weak-baseline/high-modality-gap cases. This is a protocol contribution rather than a new matcher.

2.4 Depth and Geometry Diagnostics

Depth-based reconstruction metrics are widely used in multi-view stereo and neural reconstruction [Gu et al. 2020; Wang et al. 2021; Yao et al. 2018]. However, raw UAV multispectral captures usually lack globally calibrated metric scale, and dense MVS references can fail under low texture or repeated agricultural structure. We therefore treat reference-depth as a validation-gated diagnostic rather than an unconditional benchmark: only references with valid geometric depth, normals, fusion, and support are eligible for quantitative reporting. This keeps our geometry discussion closer to repeatability/self-consistency under controlled probes than to absolute-meter accuracy claims.

3 Method

3.1 Problem Formulation

Given RGB and multispectral UAV captures, our goal is to synthesize spectral novel views and spectral indices on a common 3D support. Let $\mathcal{G}$ denote Gaussian support parameters, including position, scale, rotation, opacity, and identity, and let $\mathcal{A}_b$ denote appearance parameters for band b . UMGS separates support estimation from spectral appearance learning: it estimates $\mathcal{G}$ once from RGB, then keeps $\mathcal{G}$ fixed while learning $\mathcal{A}_b$ for each multispectral band.

3.2 Stage 1: RGB Geometry Anchor

We first train an RGB Gaussian model with the 3DGS optimization pipeline [Kerbl et al. 2023]. RGB is used as the anchor because it provides stronger texture and more stable feature matching than the spectral bands in our UAV captures. The optimized support $\mathcal{G}^*$ becomes the geometric carrier for all subsequent spectral rendering.

3.3 Stage 2: Geometry-Locked Spectral Appearance

UMGS-I (independent transfer). For each multispectral band b , we initialize from $\mathcal{G}^*$, freeze all support parameters, and optimize only $\mathcal{A}_b$. This produces independent spectral appearance models whose Gaussians remain in one-to-one correspondence.

UMGS-J (joint band-bank transfer). We keep the same frozen $\mathcal{G}^*$ but optimize band-specific appearance banks in one joint multispectral representation. Only the active bank is updated at each step, so all bands share the same support while retaining separate appearance capacity.

Geometry lock. Both variants enforce:

- fixed Gaussian count and identity;
- fixed xyz/scale/rotation/opacity;
- appearance-only parameter updates in stage 2.

These constraints define the shared support used by spectral indices. A variant that optimizes per-band xyz, scale, rotation, or opacity can still render each band independently, but its i -th Gaussian no longer denotes the same physical support across bands. Model-level ratios such as NDVI are then no longer defined on a consistent 3D support.

3.4 Raw-Scene Rectification and Baseline-Aware QA

Raw UAV scenes require spectral-to-RGB rectification before reconstruction. We estimate per-band homographies using deterministic candidate sampling and robust aggregation from feature matches. Because naive scaling baselines can be unreliable under cross-FOV and modality gaps, we use a baseline-aware QA protocol:

- **strict pass:** legacy edge+gradient improvement and geometry checks;
- **pass-with-warning:** allowed only for weak-baseline/high-modality-gap cases that satisfy match quality, stability, gradient gain, edge floor, and outlier-ratio constraints;
- **fail:** unstable geometry or insufficient matching evidence.

This protocol supports raw scenes only; aligned inputs use the same geometry-locked representation.

3.5 Products and Geometry Diagnostics

We render G/R/RE/NIR and compute NDVI, GNDVI, and NDRE under a common valid mask. When a held-out reference source passes validation, geometry can be diagnosed by relative depth agreement on probe views:

$$\delta_{rel} = \frac{D_{model} - D_{ref}}{D_{ref}}.$$

Because global metric scale is not guaranteed in raw UAV reconstructions, this diagnostic uses relative-depth ratios rather than

meter-based absolute accuracy. Failed reference sources are excluded rather than forced into the quantitative tables.

4 Experiments

4.1 Setup

Scenes. We evaluate the main paper on a 10-scene UAV-focused benchmark: 7 raw UAV scenes (raw_self, raw001–raw006) and 3 aligned aerial multispectral scenes (ms_golf, ms_lake, ms_solar). Raw scenes test rectification, QA, reconstruction, and products. Aligned scenes remove rectification confounds and are used for external common-mask comparison. Four additional non-UAV official aligned scenes are reported only in the supplement and are not part of the main 10-scene suite.

Methods. We report two UMGS variants: UMGS-I, with independent per-band transfer, and UMGS-J, with joint band-bank transfer. MS-Splatting is the external baseline on aligned aerial scenes. Scratch-MS and Unfrozen-support are representative single-band upper-bound ablations; they are not product models because they do not preserve the shared Gaussian support required by model-level multispectral fusion.

Metrics. Image quality: PSNR/SSIM/LPIPS. Spectral means average G/R/RE/NIR. Index fidelity: MAE/RMSE/PSNR/SSIM for NDVI, GNDVI, NDRE. Cost uses non-invasive GPU traces and exported Gaussian counts. Aligned external comparison uses a common-mask protocol; [†] marks UMGS renders resized to the external adapter resolution for comparison only, not distinct trained variants. We report geometry only as a validation-gated reference-depth diagnostic. Scenes whose reference mesh or camera-frame alignment failed inspection are excluded from depth tables rather than forced into an average.

4.2 Main Results on UAV Scenes

The main question is whether locking RGB geometry sacrifices spectral rendering quality. Table 1 shows that it does not: UMGS-I and UMGS-J remain near-identical on aggregate image and index quality. On the 10-scene UAV-focused suite, both reach 0.843 spectral SSIM and 0.844 index SSIM; UMGS-I has a small edge on index RMSE (0.075 vs. 0.077). We therefore use UMGS-I as the conservative primary line and treat UMGS-J as a joint-parameterization branch with comparable performance.

4.3 Index-Focused Analysis

Spectral-index synthesis is the downstream use case that motivates shared support. Table 2 breaks out NDVI, GNDVI, and NDRE. UMGS-I and UMGS-J are close on NDVI and GNDVI, while NDRE shows a larger RMSE/PSNR gap favoring UMGS-I. The joint-bank design is therefore viable but not consistently superior.

4.4 Reference-Depth Diagnostics

Table 3 reports relative-depth consistency only for reference sources that passed visual and protocol validation. The diagnostic is not an absolute geometry benchmark: it evaluates agreement with a scene-specific reference under held-out probe cameras. On the validated set, UMGS-I and UMGS-J have much lower overall relative

Table 1. Main UMGS results over the 10-scene UAV-focused suite. RGB metrics for UMGS-J use the shared UMGS-I RGB anchor. Spectral metrics average G/R/RE/NIR. Index metrics average NDVI/GNDVI/NDRE. Geometry diagnostics require validated reference sources and are therefore kept separate from this main quality table.

Method	Split	RGB PSNR	RGB SSIM	MS PSNR	MS SSIM	MS LPIPS	Idx. RMSE	Idx. SSIM
UMGS-I	Raw 7	24.39	0.840	27.40	0.850	0.136	0.065	0.869
UMGS-I	Aligned 3	25.48	0.810	26.50	0.828	0.230	0.098	0.787
UMGS-I	UAV 10	24.72	0.831	27.13	0.843	0.164	0.075	0.844
UMGS-J	Raw 7	24.39	0.840	27.37	0.850	0.136	0.067	0.869
UMGS-J	Aligned 3	25.48	0.810	26.50	0.828	0.230	0.098	0.787
UMGS-J	UAV 10	24.72	0.831	27.11	0.843	0.164	0.077	0.844

Table 2. UAV-focused 10-scene spectral-index fidelity by vegetation index.

Method	Index	MAE	RMSE	PSNR	SSIM
UMGS-I	NDVI	0.058	0.083	28.49	0.847
UMGS-I	GNDVI	0.059	0.083	28.65	0.823
UMGS-I	NDRE	0.041	0.059	31.53	0.864
UMGS-J	NDVI	0.058	0.083	28.49	0.847
UMGS-J	GNDVI	0.059	0.083	28.66	0.823
UMGS-J	NDRE	0.043	0.063	31.31	0.863

Table 3. Validation-gated reference-depth consistency. UMGS rows average the four validated reference scenes (self_m3m, raw001, raw003, and ms_golf). MS-Splatting is reported only on the available retained-self_m3m adapter; the other representative scenes only produced adapter-failure manifests.

Method / scope	Valid $\uparrow$	AbsRel $\downarrow$	Ag@1% $\uparrow$	Ag@5% $\uparrow$	Ag@10% $\uparrow$
UMGS-I / 4 scenes	0.952	0.035	0.291	0.725	0.902
UMGS-J / 4 scenes	0.952	0.035	0.291	0.725	0.902
MS-Splatting / retained self only	0.977	0.113	0.516	0.854	0.914

depth error than the available retained-self MS-Splatting adapter (AbsRel 0.035 vs. 0.113). The retained adapter has higher strict Ag@1%/Ag@5%, but it also shows a larger error tail and does not provide valid depth adapters for the other representative scenes. The accompanying 10-view depth grids show smoother UMGS error maps and more local holes or patchy deviations in the retained MS-Splatting output.

4.5 Aligned External Comparison

Aligned scenes remove the raw-rectification factor and isolate the reconstruction model. Under this common-mask setting, Table 4 shows that UMGS gives the stronger product-oriented comparison: it has lower LPIPS, matches mean SSIM/SAM, and remains within a very small margin in PSNR and spectral RMSE. Scene-level results in Table 5 are not one-sided on every scalar metric, but the official-average behavior supports UMGS as the more reliable choice for the UAV spectral-index pipeline.

Raw unaligned data test a different question: whether an external method can be applied under its retained-subset path. We keep this retained-subset evidence in the supplement as an applicability diagnostic rather than a main raw benchmark, because it does not represent the full raw protocol used by UMGS-I/UMGS-J.

Table 4. Aligned aerial common-mask comparison against MS-Splatting. Lower is better for LPIPS, 4-band RMSE, and SAM. Daggered UMGS rows are resolution-aligned comparison views, not separate training variants.

Method	PSNR	SSIM	LPIPS	4-band RMSE	SAM $^\circ$
UMGS-I †	27.01	0.853	0.215	0.0545	5.38
UMGS-J †	27.01	0.853	0.215	0.0545	5.38
MS-Splatting	27.04	0.853	0.244	0.0541	5.38

4.6 Representative Ablations

The ablations test whether geometry locking is merely conservative or actually necessary. Table 6 compares the locked UMGS variants with two upper-bound ablations on raw_self, raw002, and ms_lake. The unfrozen branch improves selected image metrics, but it is not a valid product model: on ms_lake, product construction fails the shared-support audit with an xyz mismatch of 6.84×10^{-2} . Scratch-MS diverges more strongly, including Gaussian-count mismatch across bands. These failures occur when multispectral products are assembled, showing that per-band quality gains alone are insufficient if support consistency is lost.

4.7 Cost

Finally, Table 7 reports trace-based cost indicators from AutoDL runs. UMGS-J records higher peak and average VRAM because it maintains a joint multispectral training state, while both UMGS variants keep the same exported Gaussian count by construction. These are non-invasive run traces, not a standardized FPS microbenchmark.

5 Limitations

First, aligned external comparison is scene-dependent in per-scene image metrics. UMGS improves the full official aligned average and is lower in LPIPS on the UAV aligned subset, but individual scenes can favor MS-Splatting in PSNR or spectral RMSE. We therefore emphasize protocol reliability, product readiness, and reference-depth behavior rather than claiming that every scalar metric wins on every scene.

Second, UMGS-I and UMGS-J are very close numerically. This suggests that the geometry-lock principle matters more than the specific independent or joint spectral parameterization. In the current paper, UMGS-J is best interpreted as a strong alternative branch rather than a strict replacement for UMGS-I.

Table 5. Scene-level aligned aerial common-mask comparison for UMGS-I[†] and MS-Splatting. PSNR/SSIM/LPIPS average G/R/RE/NIR; 4-band RMSE and SAM evaluate the spectral vector.

Scene	UMGS PSNR	MMS PSNR	UMGS SSIM	MMS SSIM	UMGS LPIPS	MMS LPIPS	UMGS RMSE	MMS RMSE	UMGS SAM	MMS SAM
ms_golf	31.33	32.12	0.928	0.935	0.176	0.234	0.0300	0.0273	2.77	3.11
ms_lake	22.02	22.79	0.740	0.731	0.350	0.374	0.0887	0.0817	8.84	7.88
ms_solar	27.72	26.21	0.893	0.893	0.117	0.123	0.0448	0.0532	4.52	5.16

Table 6. Representative 3-scene ablation summary on raw_self, raw002, and ms_lake. Spectral metrics average G/R/RE/NIR; index metrics average NDVI/GNDVI/NDRE. Scratch-MS and Unfrozen-support are single-band upper-bound ablations, not valid product models, because their shared-support audits fail.

Method	MS PSNR	MS SSIM	MS LPIPS	Idx. RMSE	Shared	Products
UMGS-I	25.35	0.786	0.228	0.100	yes	yes
UMGS-J	25.33	0.786	0.228	0.100	yes	yes
Scratch-MS	25.88	0.780	0.289	0.116	no	invalid
Unfrozen-support	26.27	0.794	0.218	0.105	no	invalid

Table 7. Trace-based cost indicators from AutoDL runs. VRAM, utilization, and power are sample-weighted over non-invasive GPU traces; Gaussian counts are exported PLY vertex counts for evaluation models.

Method	Peak GB	Avg. GB	Util. %	Power W	Gauss. M
UMGS-I	27.45	12.09	61.4	226.4	4.63
UMGS-J	35.04	18.70	77.0	274.9	4.63

Third, our geometry diagnostics are validation-gated reference-depth checks, not dense metric ground truth. We report only scenes whose reference mesh and camera frame pass validation, and we exclude collapsed dense-MVS or misaligned reference attempts from quantitative tables. This conservative policy avoids reporting invalid reference geometry, but it also means the current paper does not replace LiDAR-grade absolute geometry benchmarks.

Fourth, cost reporting is intentionally non-invasive. The reported VRAM/utilization/power/size statistics are traceable AutoDL run records, but a dedicated runtime microbenchmark (e.g., standardized per-view FPS across methods and resolutions) is still future work.

6 Conclusion

We presented UMGS, a geometry-locked multispectral 3D Gaussian Splatting framework for UAV spectral-index synthesis. UMGS anchors geometry in RGB and learns spectral appearance on fixed support, so cross-band products are formed on consistent Gaussian topology rather than loosely coupled per-band reconstructions.

The experiments support three conclusions. First, RGB-anchored geometry locking is the central design choice: it preserves support consistency even when weaker spectral bands or unfrozen per-band optimization can improve isolated image metrics. Second, the same principle remains stable under two parameterizations. UMGS-I is the conservative primary line, while UMGS-J provides a viable joint branch with comparable behavior. Third, against the available multispectral Gaussian baseline, UMGS provides stronger overall official aligned comparison behavior and better validation-gated reference-depth consistency by the primary relative-error diagnostics.

The broader implication is that UAV multispectral reconstruction should treat shared 3D support as a first-order constraint for

spectral-index synthesis, not as a secondary check after single-band rendering optimization.

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